

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-8574
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs	Yes	PEP8, 4-space indent
2	Proper File Structure	Files and folders are	Yes	/templates, /routes, /models
3	Meaningful Variable Names	Variables reflect their	Yes	user_data, template_id etc
4	Function / Method Names	Functions are descriptively	Yes	validate_input(),
5	Code Comments	Inline and block comments explain	Yes	Docstrings added for AI funcs
6	Modular Design	Code is split into reusable	Yes	Auth, AI, PDF service
7	No Redundant Code	Duplicate or unused code is	Yes	Cleaned, DRY followed
8	Error Handling	Exceptions and errors handled	Yes	Try-except, JWT expiry handled

Reusable Components / Modules:

S.No	Component Name	Language	Where Reused	Reusability Level
1	Auth Module	Python/Flask	Login, Registration	High
2	Template Engine	Jinja2	Rental, NDA, Employmen	High
3	AI Generation Service	LangChain, LLM	All doc types - RAG	High
4	PDF Export Service	PyMuPDF	Final download, storag	High
5	Validation Service	Pydantic, spaCy	Input validation	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured MVC, separate layers
Readability	5	Clean code, PEP8 compliant
Reusability	5	Modular services reused across 15+ types
Documentation / Comments	4	Docstrings, README with setup
Overall Score	5	Production-ready, maintainable